Assignment-01

1.

(i) Increasing intervals

$$-4 < x < -2, 1 < x < 4$$

Decreasing intervals

$$x < -4, -2 < x < 1, x > 4$$

(ii) Critical points $x = -4, -2, 1, 4$

(iii) $f_{min}(-4) = 0$, $f_{max}(-2) = 11$,
$f_{min}(1) = -5$, and $f_{max}(4) = 27$

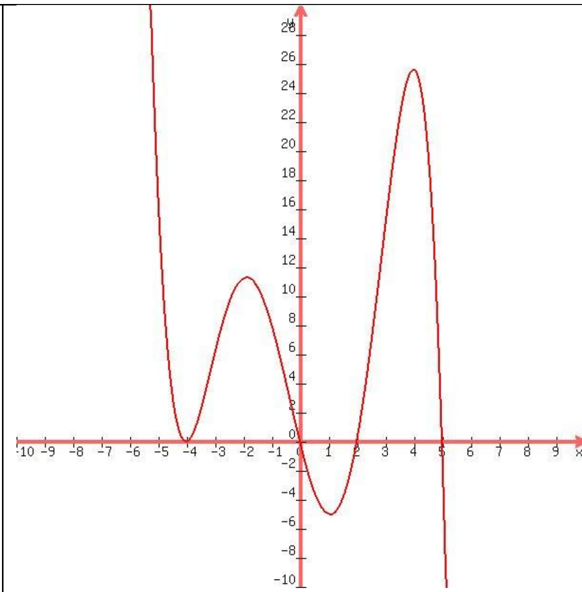
(iv) Concave up intervals

$$x < -3, -0.5 < x < 2.5$$

Concave down intervals

$$-3 < x < -0.5, x > 2.5$$

(v) Inflection points $x = -3, -0.5, 2.5$



2.

(i) $f'(x) > 0$ (increasing) intervals

$$-3 < x < -2, x > 1$$

$f'(x) < 0$ ($d$ecreasing) intervals

$$x < -3, -2 < x < 1$$

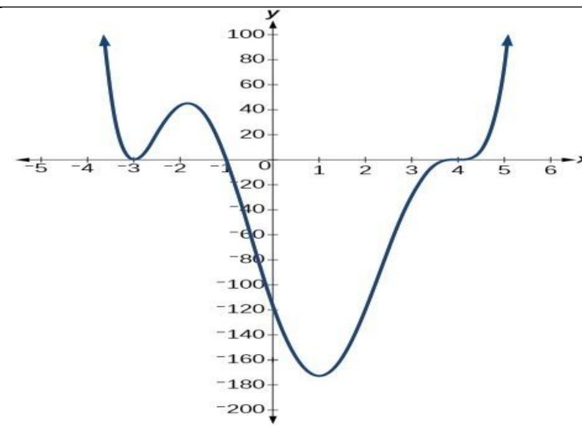
$f''(x) > 0$ (concave up) intervals

$$x < -2.5, -0.5 < x < 2.5, x > 4$$

$f''(x) < 0$ (concave down) intervals

$$-2.5 < x < -0.5, 2.5 < x < 4$$

(ii) Critical points $x = -3, -2, 1, 4$
Inflection points $x = -2.5, -0.5, 2.5, 4$



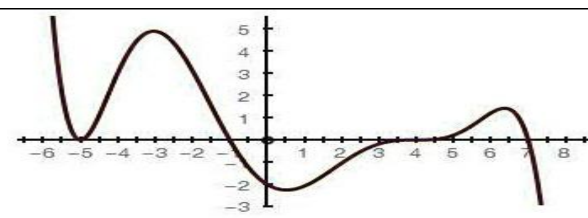
3.

$f'(x) = 0$ (critical points) at

$$x = -5, -3.5, -0.5, 4, 6.5$$

$f'(x) = 0$ (inflection points) at

$$x = -4.25, -1.5, 2.25, 4, 5.25$$

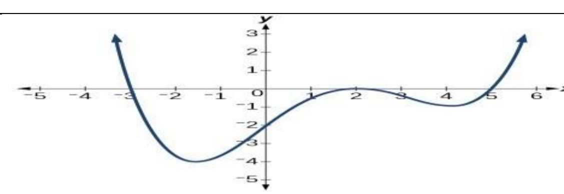


4.

$x = -3$ is a root of odd multiplicity

$x = 2$ is a root of even multiplicity

$x = 5$ is a root of odd multiplicity



Assignment-01

5.	$f(x) = 2x^3 + 3x^2 - 36x$ $f'(x) = 6x^2 + 6x - 36 = 6(x^2 + x - 6) = 6(x-2)(x+3)$ $f'(x) = 0 \rightarrow x = -3, 2$ (critical points) Increasing ($f'(x) > 0$) intervals $-\infty < x < -3, 2 < x < \infty$ Decreasing ($f'(x) < 0$) intervals $-3 < x < 2$ $f_{max}(-3) = 81$, $f_{min}(2) = -44$ $f''(x) = 12x + 6$ $f''(-3) = -30 < 0$ (maximum confirmation) $f''(2) = 30 > 0$ (minimum confirmation)

6.	Follow the Exercise no. 53 and 54 of Chapter 4.2.